Multimedia Signal Processing - 2nd Module

February, 10 2016

1. **Question (8 pts)**

   Consider two discrete memoryless sources X and Y that emit symbols from the alphabets $\{a_1, a_2\}$ and $\{b_1, b_2, b_3\}$, respectively. The joint p.m.f. is

	b_1	b_2	b_3
a_1	0.1	0.4	0.1
a_2	0.15	0.1	0.15

   (a) Design the Huffman code to encode single symbols of Y.

   (b) Compute: (i) the average code length of the obtained Huffman code; (ii) the average code length of the Shannon code, and; (iii) the theoretical bound.

   (c) Design the Huffman code that jointly encodes one symbol of X and one symbol of Y.

   (d) Compute again: (i) the average code length of the obtained Huffman code; (ii) the average code length of the Shannon code, and; (iii) the theoretical bound.

   (e) Compare and comment the results obtained in point (b) and point (d).

   (f) Compute the mutual information $I(X;Y)$.

2. **Question (5 pts)**

   (a) Illustrate the block diagram of a closed-loop DPCM coder and explain when it is useful.

   (b) Discuss the difference between a closed-loop and open-loop implementation.

3. **Question (5 pts)**

   (a) Describe the source-system model used to synthesize speech signals.

   (b) Explain how a linear predictive vocoder exploits this scheme, and which are the main parameters sent from the encoder to the decoder.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider a video coding scheme:

   - [T] [F] Motion estimation is performed in the KLT domain.
   - [T] [F] Motion vectors are estimated at the encoder and transmitted to the decoder.
   - [T] [F] Motion vectors can be estimated with different algorithms, all giving the same result.
   - [T] [F] Motion estimation can be applied also on blocks whose size is different from 8×8 pixels.

   (b) Consider perceptual audio coding schemes:

   - [T] [F] Both tonal and noise maskers are considered by the MPEG-1 psychoacoustic model.
   - [T] [F] MDCT transform is not critically sampled as M coefficients are generated from each $2M$ samples.

   (c) Consider an image coding scheme:

   - [T] [F] The YCbCr color-space does not provide any advantage over the RGB one.
   - [T] [F] The 2D DFT transform is separable.

   (d) Consider a continuous random variable X with uniform p.d.f. and standard deviation σ_X:

   - [T] [F] For quantization, the use of compander-expander is preferable to the use of a scalar uniform quantizer.
   - [T] [F] According to rate-distortion theory, $R(D) = 0$ for $D = \sigma_X^2$.

5. **Question (Matlab - 10pts)**

   (a) Read the image *image1.png*. Convert it from the RGB to the YCbCr color space according to the transformation

$$\begin{bmatrix} x_Y \\ x_{Cb} \\ x_{Cr} \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.5 \\ 0.5 & -0.419 & -0.0813 \end{bmatrix} \begin{bmatrix} x_R \\ x_G \\ x_B \end{bmatrix}$$

   (b) Considering only the Y component, implement a block-wise transform coding scheme based on 2D DCT. Repeat the coding step six times using each time a different block-size and quantization-step: (i) 8×8 and 16×16 pixels blocks; (ii) quantization steps $\Delta \in \{2, 11, 20\}$, paying attention to use a midtread quantizer.

   (c) Plot two rate-distortion (RD) curves, i.e., one for each different block size. Use PSNR as distortion metric and assume you are able to encode quantized coefficients with the minimum amount of bits theoretically possible.

   (d) Repeat the same steps for the file *image2.png*. Compare RD curves of *image1.png* with those of *image2.png*. Comment what you see (hint: have a look at the histograms of DCT coefficients before and after quantization).